Frequency Analysis

ECSE 313 Group 16 Lab 3 Report

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Contents

1 Background **2**
1.1 Synthesis of Periodic Signals 2
1.2 Magnitude and Phase of Discrete-Time Systems 3

2 Continuous-Time Frequency Analysis **4**
2.1 Synthesis of Periodic Signals 4
2.2 Modulation Property 4
2.3 System Analysis 6

3 Discrete-Time Frequency Analysis **7**
3.1 Discrete-Time Fourier Transform 7
3.2 System Analysis 8

1 Background

1.1 Synthesis of Periodic Signals

Using the Fourier series expansion in the form

$$s(t) = a_0 + \sum_{k=1}^{\infty} A_k sin(2\pi k f_0 t + \theta_k) \tag{1}$$

for the two equations

$$s_1(t) = rect(t - \frac{1}{2}) \tag{2}$$

$$s_2(t) = rect(2t) - \frac{1}{2} \tag{3}$$

The piecewise function in Equation 2 is only 1 from $t = 0$ to $t = 1$ and 0 elsewhere. Therefore, the function becomes:

$$f_1(t) = \begin{cases} 1, & 0 < t < 1 \\ 0, & 1 < t < 2 \end{cases} \tag{4}$$

Using a period ($T_0 = 2$) from $t \in [0, 2]$ and $f_0 = \frac{1}{T} = \frac{1}{2}$, the a_0 value of Equation 2 becomes

$$a_0 = \frac{1}{T}\int_0^T f_1(t)dt = \frac{1}{2}\int_0^1 1dt = \frac{1}{2} \tag{5}$$

The fundamental frequency of the function is $w_0 = \frac{2\pi}{T} = \pi$. Moreover, a_k is

$$a_k = \frac{2}{T}\int_0^T f_1(t)cos(kw_0 t)dt = \int_0^1 cos(k\pi t)dt = \frac{sin(k\pi)}{k\pi} = 0 \tag{6}$$

a_k is zero because the sine function is zero for every integer value of k. The odd part of the signal, b_k, is defined as

$$b_k = \frac{2}{T}\int_0^T f_1(t)sin(kw_0 t)dt = \int_0^1 sin(k\pi t)dt = \frac{1 - cos(k\pi)}{k\pi} \tag{7}$$

In Equation 7, when k is even, $b_k = \frac{1-1}{k\pi} = 0$. When k is odd, $b_k = \frac{1-(-1)}{k\pi} = \frac{2}{k\pi}$. Moreover, $A_k = b_k$ and $\theta_k = 0$. Thus, Equation 2 becomes

$$s_1(t) = \frac{1}{2} + \sum_{k=1,3,5,...}^{\infty} \frac{2}{k\pi} sin(\pi k t) \tag{8}$$

The piecewise function in Equation 3 is only $1 - \frac{1}{2} = \frac{1}{2}$ from $t = -\frac{1}{4}$ to $t = \frac{1}{4}$ and $-\frac{1}{2}$ elsewhere. Therefore, the function becomes:

$$f_2(t) = \begin{cases} \frac{1}{2}, & |t| < \frac{1}{4} \\ -\frac{1}{2}, & \frac{1}{4} < |t| < \frac{1}{2} \end{cases} \tag{9}$$

Since $f_2(t)$ is an even function (symmetric across the y-axis), $b_k = 0$. Using a period ($T_0 = 1$) from $t \in [-\frac{1}{2}, \frac{1}{2}]$, the a_0 value of Equation 3 becomes

$$a_0 = \frac{1}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f_2(t)dt = \int_{-\frac{1}{2}}^{-\frac{1}{4}} -\frac{1}{2}dt + \int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{1}{2}dt + \int_{\frac{1}{4}}^{\frac{1}{2}} -\frac{1}{2}dt = 0 \tag{10}$$

The fundamental frequency of the function is $w_0 = \frac{2\pi}{T} = 2\pi$ and $f_0 = \frac{1}{T} = 1$. Moreover, a_k is

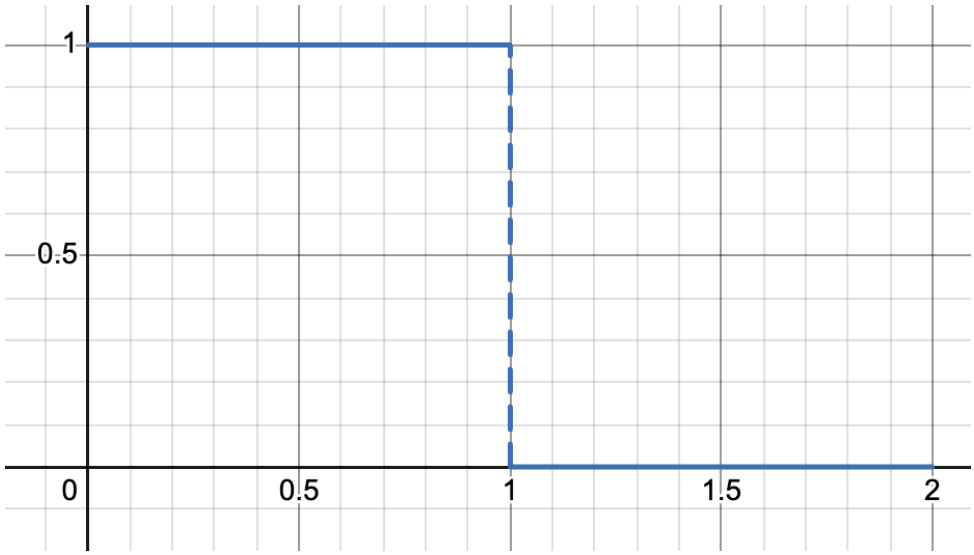

Figure 1: Sketch of Equation 2 from $[0, T_0]$

$$a_k = \frac{2}{T}\int_{-\frac{T}{2}}^{\frac{T}{2}} f_2(t)cos(kw_0t)dt = 2\int_{-\frac{1}{2}}^{-\frac{1}{4}} -\frac{1}{2}cos(2\pi kt) + 2\int_{-\frac{1}{4}}^{\frac{1}{4}} \frac{1}{2}cos(2\pi kt) + 2\int_{\frac{1}{4}}^{\frac{1}{2}} -\frac{1}{2}cos(2\pi kt) \quad (11)$$

$$a_k = -\left[\frac{sin(2\pi kt)}{2\pi k}\right]_{-\frac{1}{2}}^{-\frac{1}{4}} + \left[\frac{sin(2\pi kt)}{2\pi k}\right]_{-\frac{1}{4}}^{\frac{1}{4}} - \left[\frac{sin(2\pi kt)}{2\pi k}\right]_{\frac{1}{4}}^{\frac{1}{2}} \quad (12)$$

$$a_k = \frac{4sin(\frac{\pi k}{2}) - 2sin(\pi k)}{2\pi k} = \frac{2sin(\frac{k\pi}{2})}{k\pi} \quad (13)$$

$A_k = a_k$. When k is even, $A_k = 0$. For $k = 1, 5, \ldots,$ $A_k = \frac{2}{k\pi}$, and for $k = 3, 7, \ldots,$ $A_k = -\frac{2}{k\pi}$. Additionally, converting the cosine function to sine function using the identity $cos(\theta) = sin(\theta + \pi/2)$, the θ values become $\theta = \frac{\pi}{2}$. Since A_k alternates by sign every other odd value, the θ value is positive for $k = 1, 5, \ldots$ and negative for $k = 3, 7, \ldots$. Thus, Equation 3 becomes

$$s_2(t) = \sum_{k=1,3,5,\ldots}^{\infty} \frac{2}{k\pi} sin(2\pi kt + \theta) \quad (14)$$

where $\theta = \frac{\pi}{2}$ for $k = 1, 5, \ldots$ and $\theta = -\frac{\pi}{2}$ for $k = 3, 7, \ldots$

1.2 Magnitude and Phase of Discrete-Time Systems

For the following function:

$$y(n) = 0.9y(n-1) + 0.3x(n) + 0.24x(n-1) \quad (15)$$

By setting $x(n) = \delta(n)$, the impulse reponse is found through setting $n = 0$.

$$y(0) = 0.9y(-1) + 0.3\delta(0) + 0.24\delta(-1) = 0.9(0) + 0.3(1) + 0.24(0) = 0.3 \quad (16)$$

$$y(1) = 0.9y(0) + 0.3\delta(1) + 0.24\delta(0) = 0.9(0.3) + 0.3(0) + 0.24(1) = 0.51 \quad (17)$$

For $n > 1$, Equation 15 is just $y(n) = 0.9y(n-1)$. Therefore the impulse is

$$h(n) = 0.3\delta(n) + 0.51(0.9)^{n-1}u(n-1) \quad (18)$$

The system diagram of Equation 15 is shown in Figure 3.

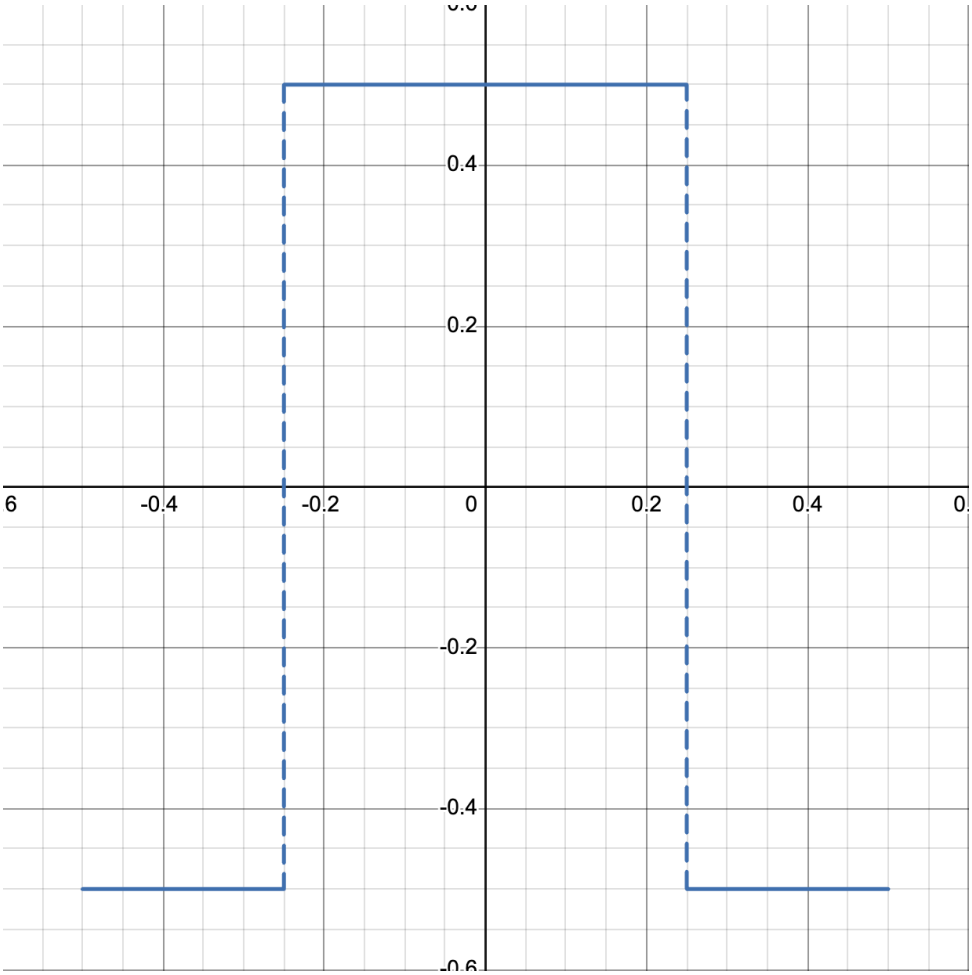

Figure 2: Sketch of Equation 2 from $[-\frac{T_0}{2}, \frac{T_0}{2}]$

The Z-transform of Equation 15 is found using the equation $x(n-k) \leftrightarrow z^{-k}X(z)$

$$Y(z) = 0.9z^{-1}Y(z) + 0.3X(z) + 0.24z^{-1}X(z) \tag{19}$$

The transfer function of Equation 15 using Equation 19 is

$$(1 - 0.9z^{-1})Y(z) = (0.3 + 0.24z^{-1})X(z) \tag{20}$$

$$H(z) = \frac{0.3 + 0.24z^{-1}}{1 - 0.9z^{-1}} \tag{21}$$

The magnitude and phase responses of Equation 21 are plotted in Figure 4

2 Continuous-Time Frequency Analysis

2.1 Synthesis of Periodic Signals

Figure 7 has the highest period while Figure 6 has the lowest period. The frequency in Hertz is about the same in the Spectral Density plots.

2.2 Modulation Property

Changing the modulating frequency results in a frequency shift from 10Hz to 15Hz. The spectrum has a comb structure because the signal being modulated, triangle pulse, is periodic. Periodic signals can be represented by impulses, and the distance between the impulses is found through $f = \frac{1}{T}$. If the

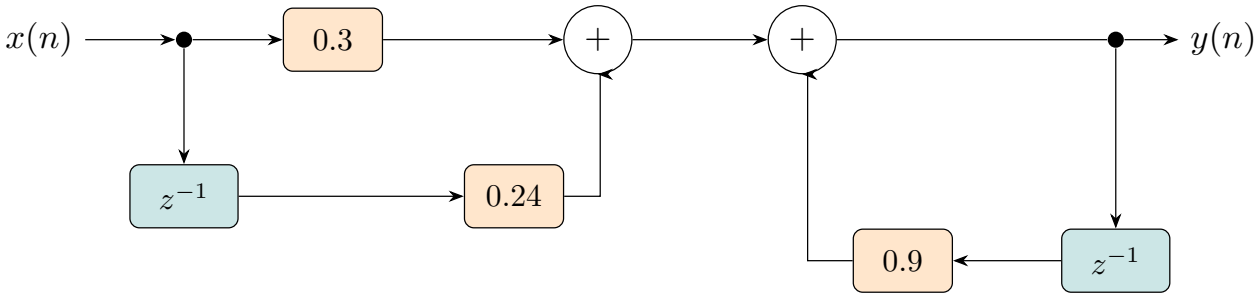

Figure 3: Direct Form I: $y(n) = 0.9\,y(n-1) + 0.3\,x(n) + 0.24\,x(n-1)$

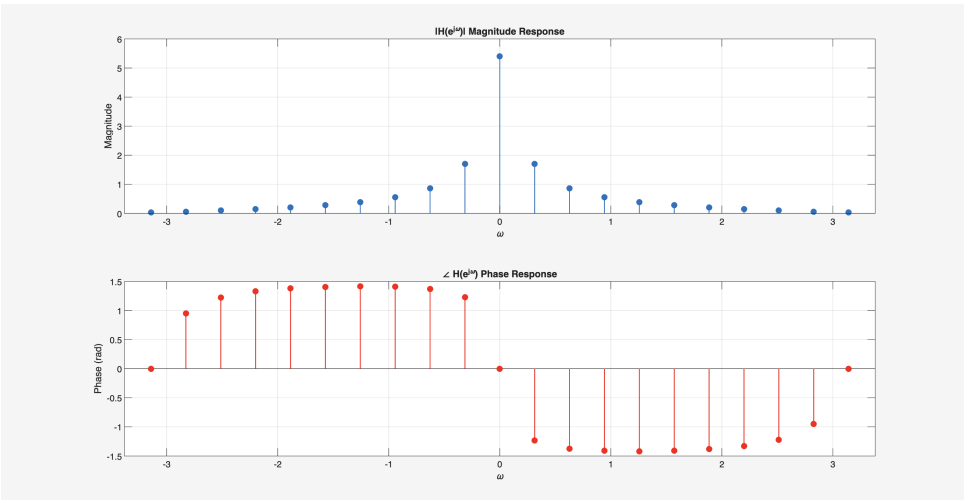

Figure 4: Discrete-Time Magnitude and Phase Responses of Equation 21

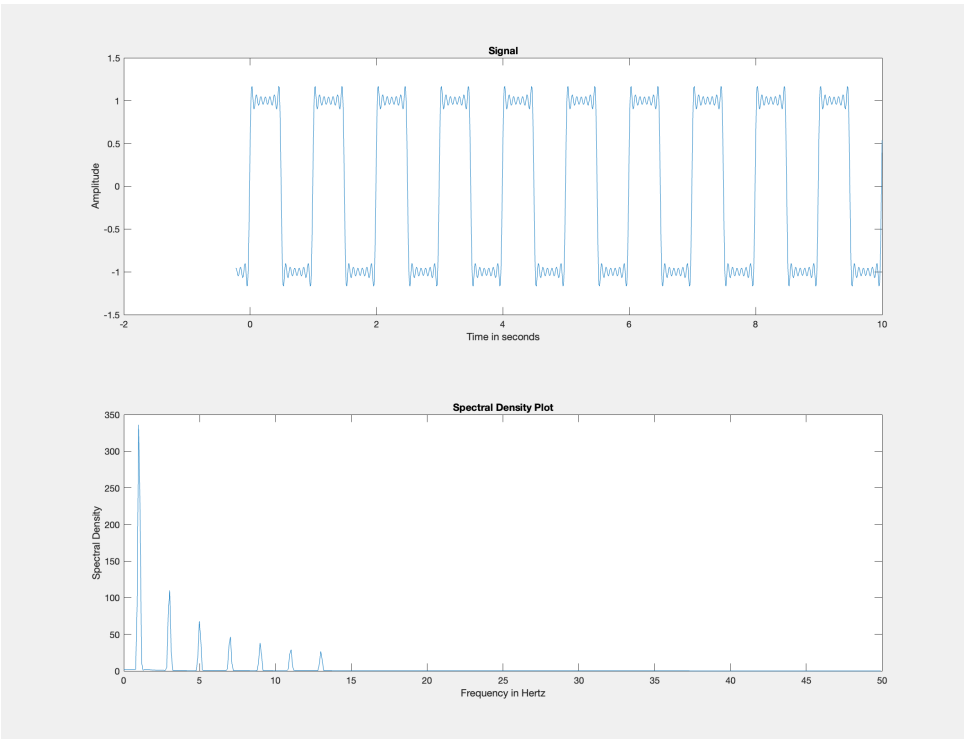

Figure 5: Preconfigured Fourier Representation of Signal

period of the triangle pulse were to increase toward infinity, the triangle pulse would no longer have

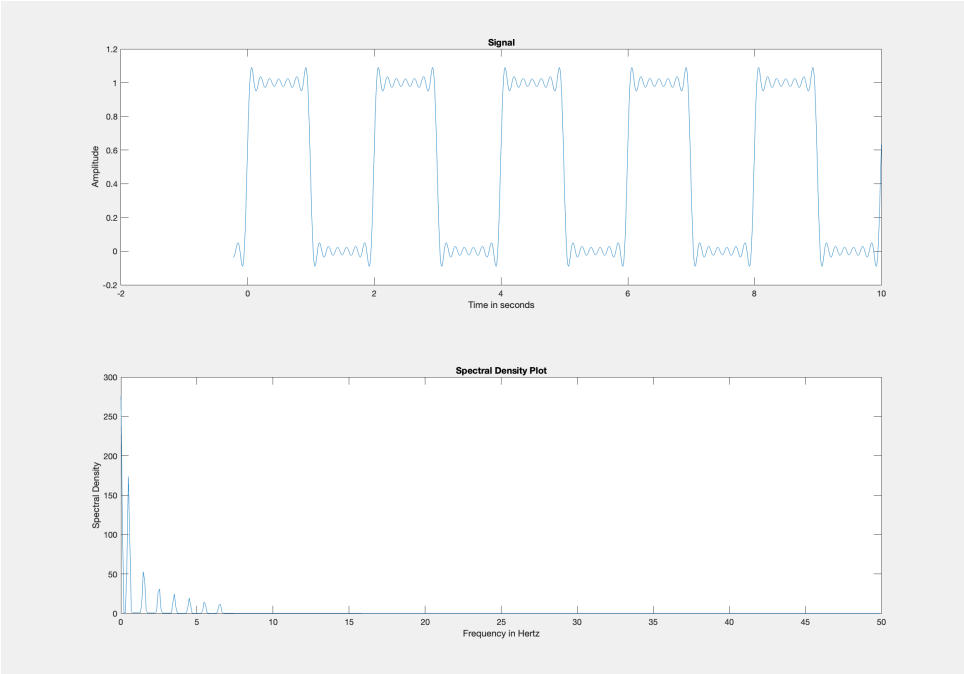

Figure 6: Preconfigured Fourier Representation of Signal

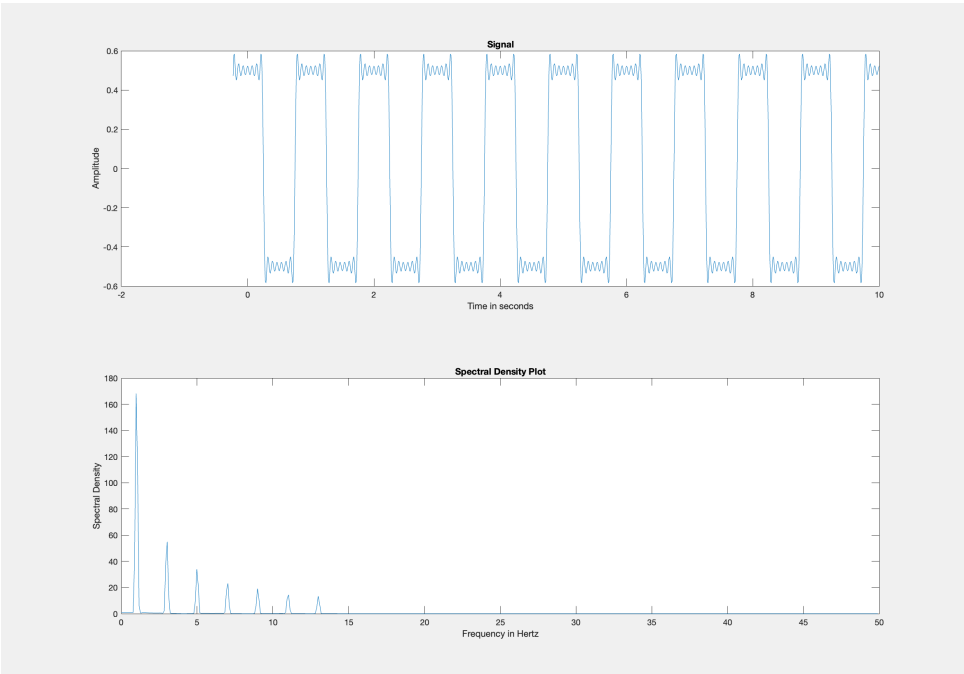

Figure 7: Preconfigured Fourier Representation of Signal

a period. Therefore, the distances between the impulses become small, and the function theoretically remains continuous.

2.3 System Analysis

The Network Analyzer is more powerful and gives you everything at once, but is more complex

The Unit Step method is simpler but differentiation is sensitive to noise, so the impulse response will be less clean — more jaggedness in that plot compared to the Network Analyzer version.

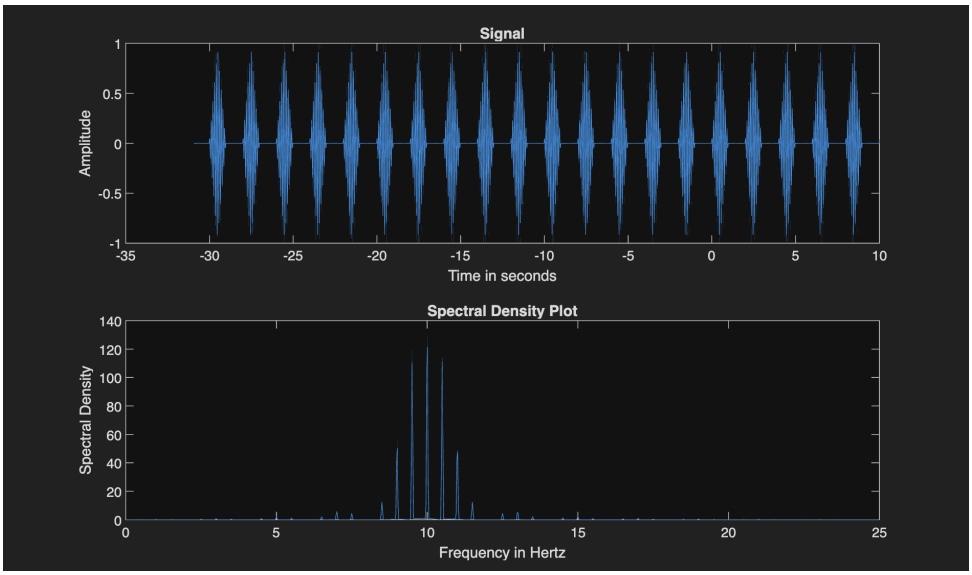

Figure 8: Spectrum Analyzer of 2 sec Triangle Pulse at 10 Hz

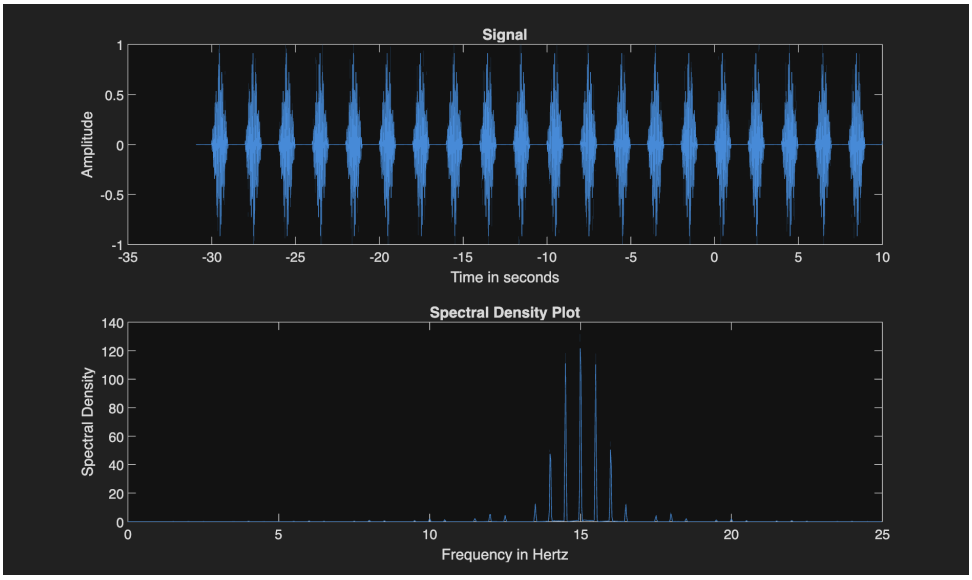

Figure 9: Spectrum Analyzer of 2 sec Triangle Pulse at 15 Hz

3 Discrete-Time Frequency Analysis

3.1 Discrete-Time Fourier Transform

dtft.m

```
% Purpose: Computes the Discrete-Time Fourier Transform (DTFT) of a Signal
% Input: (x) - vector representing the time-domain
%        (n0) - time index corresponding to 1st element in vector x
%        (dw) - spacing between samples
% Output: (X) - Values corresponding to DTFT of signal from -pi to pi

function X = dtft(x, n0, dw)
    w = -pi:dw:pi;
    N = length(x);
    n = n0:(n0 + N - 1);
    X = x * exp(-1j * n' * w);
end
```

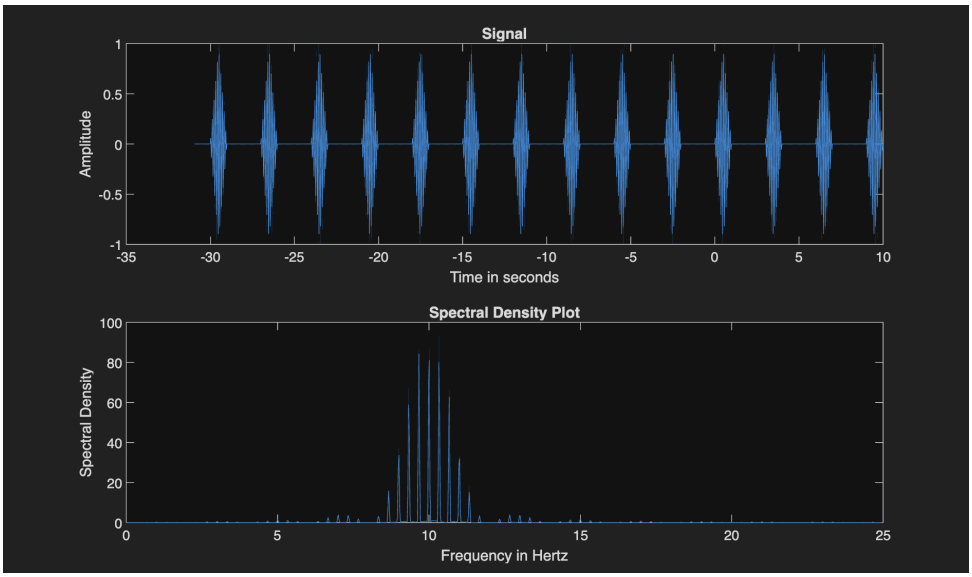

Figure 10: Spectrum Analyzer of 3 sec Triangle Pulse at 10 Hz

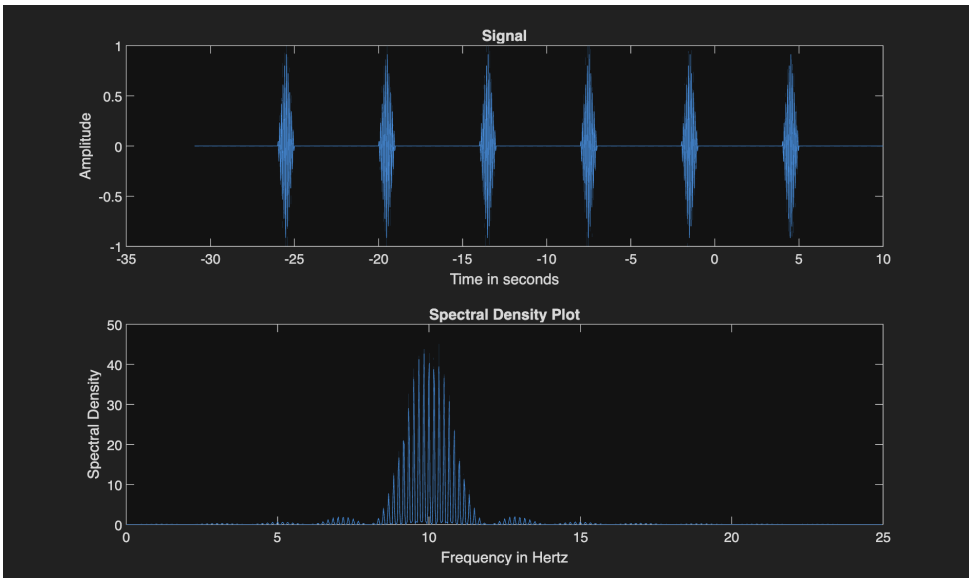

Figure 11: Spectrum Analyzer of 6 sec Triangle Pulse at 10 Hz

3.2 System Analysis

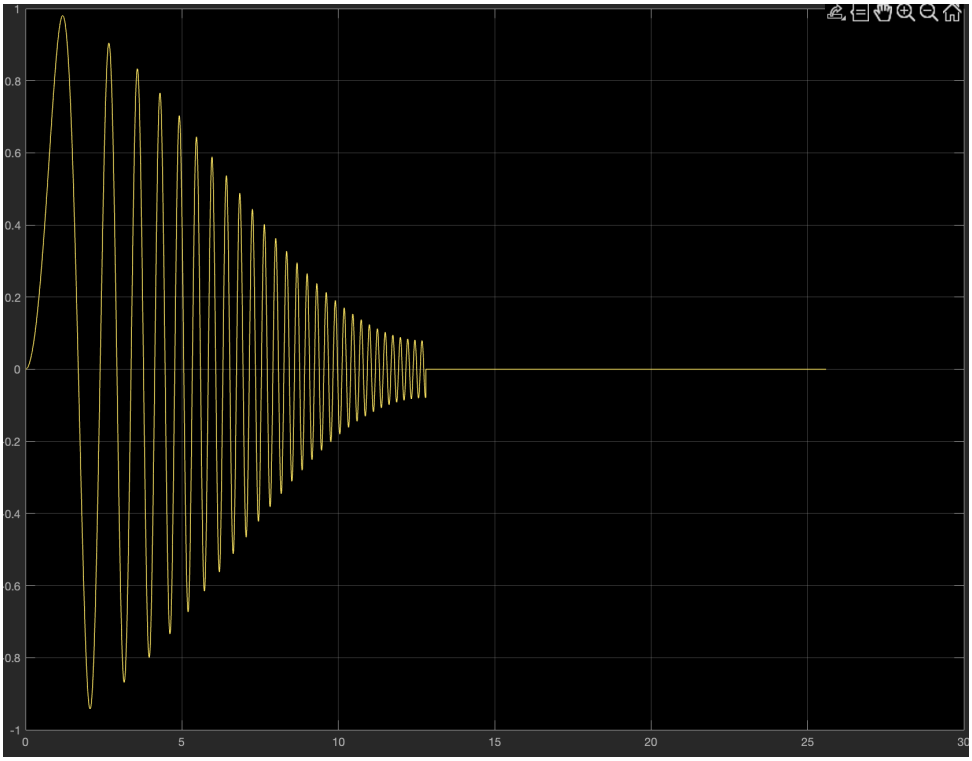

Figure 12: Output of Network Analyzer

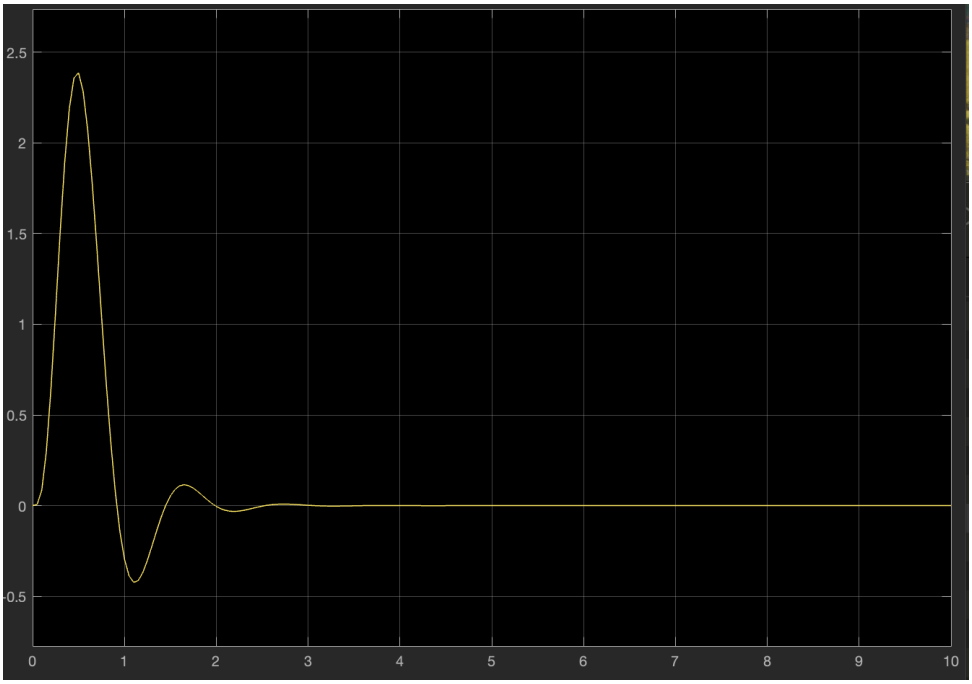

Figure 13: Output of Network Analyzer

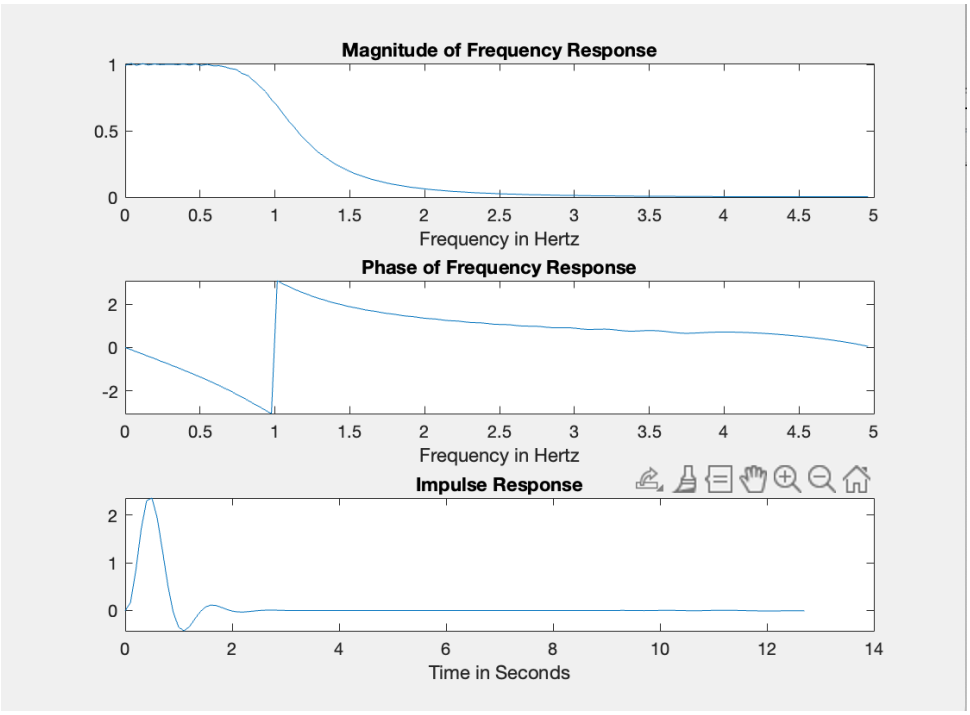

Figure 14: Magnitude, Phase, and Impulse Using Unit Step

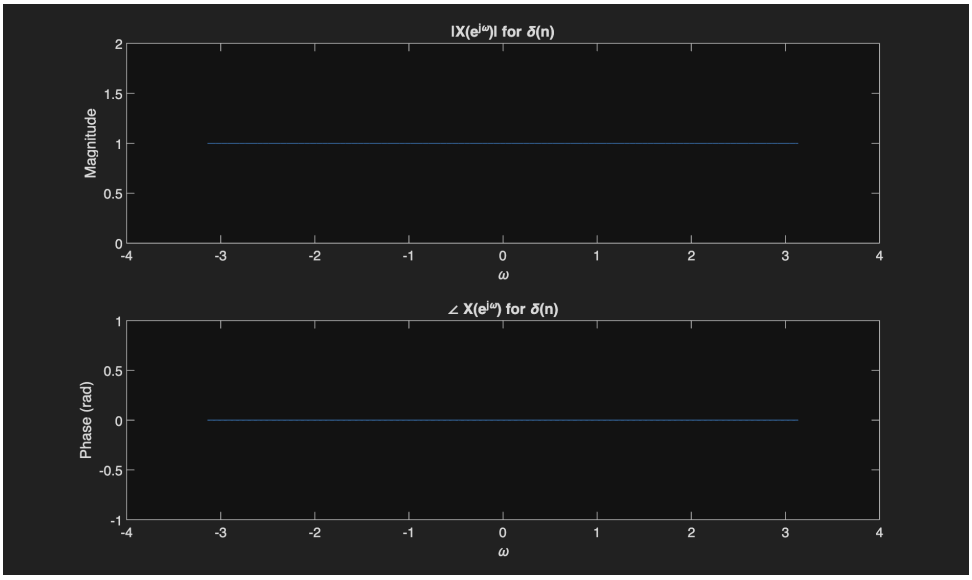

Figure 15: Magnitude and Phase Responses of $X(e^{jw})$

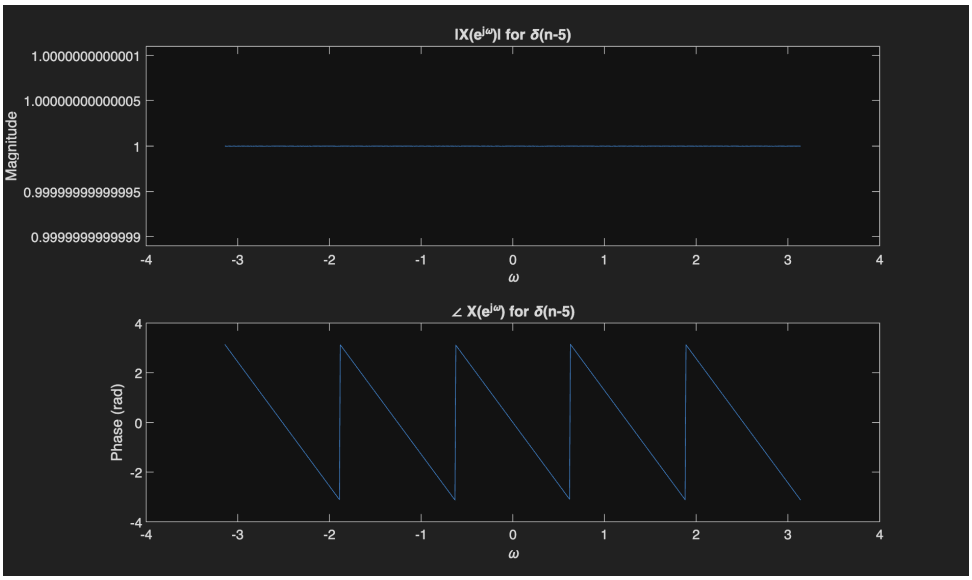

Figure 16: Magnitude and Phase Responses of $X(e^{jw})$

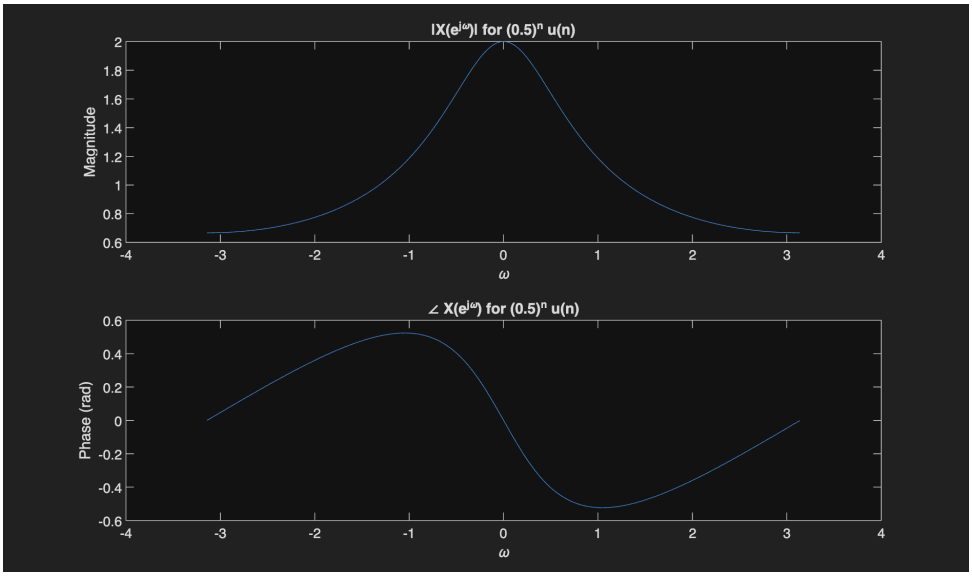

Figure 17: Magnitude and Phase Responses of $X(e^{jw})$

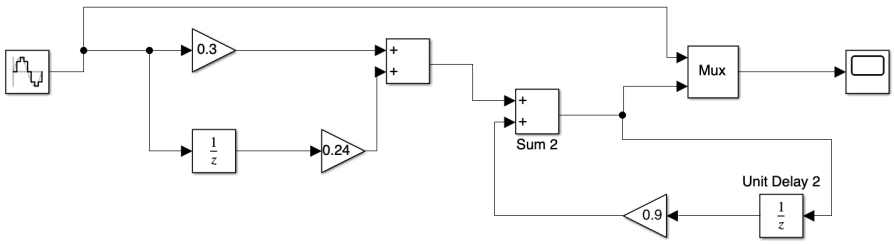

Figure 18: Completed Block Diagram of Discrete-Time System Analysis

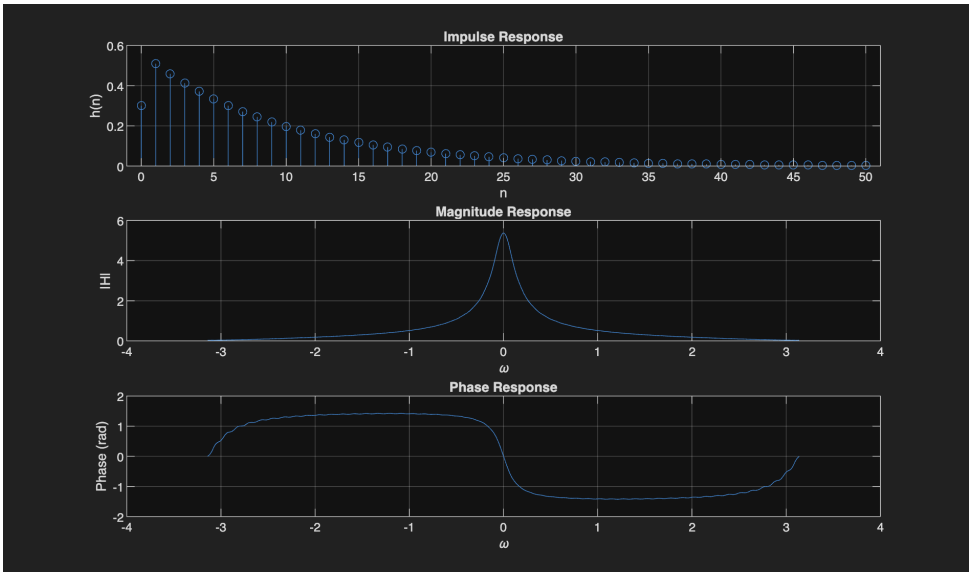

Figure 19: Impulse Response, Magnitude, and Phase of the Frequency Response

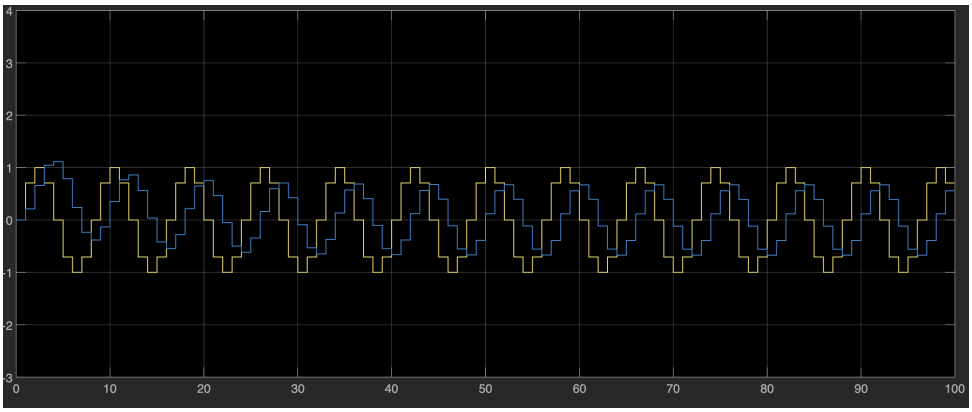

Figure 20: Sine Wave at $w = \pi/4$

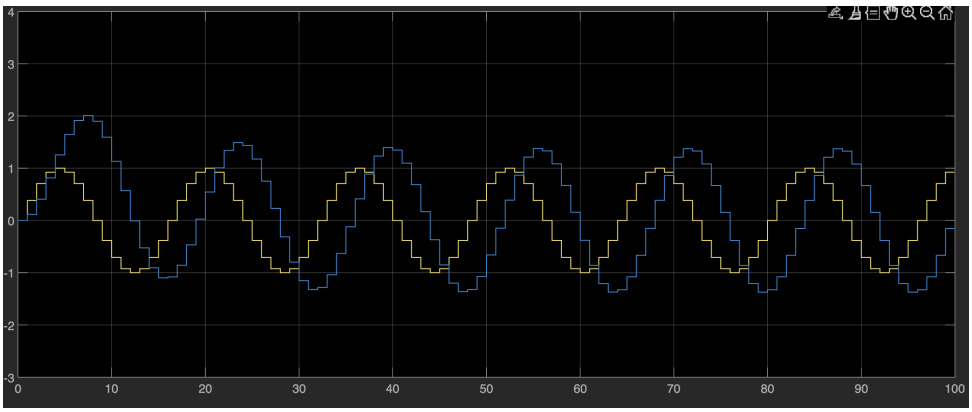

Figure 21: Sine Wave at $w = \pi/8$

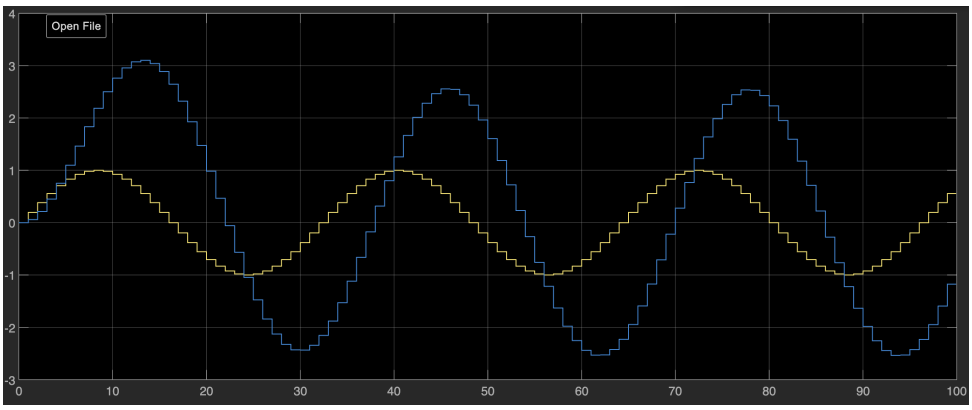

Figure 22: Sine Wave at $w = \pi/16$

Frequency	Theoretical |H|
pi/16	2.5452
pi/8	1.3816
pi/4	0.6814

Figure 23: Table of Theoretical Values with Frequencies